01. AgentFlow

Description:

AI-Powered Workflow Automation Platform

Concept: Users create automated workflows by connecting different actions, triggers, and AI agents. Similar in concept to visual automation platforms.

Features

- Drag-and-drop workflow builder.
- AI-powered workflow generation from text.
- Scheduled and event-based triggers.
- Connect multiple APIs.
- Workflow execution history.
- Error handling and retry mechanisms.
- User-created workflow templates.

Development stack

- Frontend: React, Tailwind CSS, React Flow.
- Backend: Node.js, Express.js.
- Database: MongoDB.
- Background jobs: BullMQ + Redis.
- AI: LLM API.
- Real-time: Socket.io.

Database models: User, Workflow, Node, Execution, Integration, Trigger.

Development phases

1. Build authentication and dashboard.
2. Create visual workflow editor.
3. Develop workflow execution engine.
4. Integrate external APIs.
5. Add AI workflow generation.
6. Add execution monitoring and deployment.

Placement value: API integration, asynchronous jobs, system design, and AI.

